

SHUAICHENG (ALLEN) TONG

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EDUCATION

Georgia Institute of Technology

August 2024 - Present

Ph.D. in Machine Learning, MS in Computer Science

Advisor: Pascal Van Hentenryck

Research Area: Machine Learning for Combinatorial Optimization

University of California, Los Angeles

October 2020 - June 2024

B.S. in Mathematics of Computation

Minor in Data Science Engineering

GPA: 3.9.

PUBLICATIONS

- X. Li et al. (including **S. Tong**). *SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks*, NeurIPS 2026.
- **S. Tong** and P. Van Hentenryck. *Improving Stability and Economic Operation in Transmission Systems through Volt/VAR Optimization*, The 60th Hawaii International Conference on System Sciences (HICSS-60).
- **S. Tong**, M. A. Boateng, M. Tanneau, and P. Van Hentenryck. *Volt/VAR Optimization in Transmission Networks with Discrete-Control Devices*, International Power Systems Computation Conference (PSCC 2026), and special issue of Electric Power Systems Research (EPSR).
- T. Ye, **S. Tong**, C. Guan, B. Basciftci, and P. Van Hentenryck. *Learning Optimization Proxies for Sequential Contextual Stochastic Programs: An Order Fulfillment Application*, under consideration at Manufacturing & Service Operations Management (MSOM).
- T. Ye, **S. Tong**, C. Guan, B. Basciftci, and P. Van Hentenryck. *Deep Learning-Driven Contextual Stochastic Optimization for Real-Time Order Fulfillment*, NeurIPS 2025 Workshop MLxOR.
- L. Liu, **S. Tong**, and L. Zhao. *Training Implicit Networks for Image Deblurring using Jacobian-Free Backpropagation*, SIAM Undergraduate Research Online.

EXPERIENCES

AI Institute for Advances in Optimization (AI4OPT), Georgia Tech School of ISyE

August 2024 - Present

Graduate Research Assistant

- Benchmarked LLM agent skills for science and engineering problems, measuring their impact on agent reasoning, task success, and generalization.
- Applied machine learning (ML) and mathematical optimization to solve large-scale decision-making under uncertainty in energy and supply chain systems.
- Built modeling, simulation, and benchmarking workflows to evaluate algorithm performance, scalability, and feasibility.

Intact Financial Corporation - Data Lab

June 2023 – September 2023

Data Scientist Intern

- Built Python ETL and feature integration pipelines that expanded ML model coverage by 20% for risk assessment and insurance pricing.
- Optimized automated data workflows for production quoting, reducing average quote time by 6 minutes and improving user experience.

- Prototyped an LLM-assisted questionnaire system to improve customer risk profile accuracy for downstream modeling and customer segmentation.
- Evaluated model lift with actuaries and translated predictive insights into pricing decisions.

Emory University Math Department
Undergraduate Research Assistant

May 2022 – July 2022

- Researched implicit neural networks for image processing, focusing on scalable and memory-efficient model training.
- Developed and benchmarked Jacobian-free Backpropagation (JFB) against classical optimization, feed-forward neural networks, and deep unrolling baselines.
- Analyzed model robustness, and communicated findings through a manuscript and project website.

UCLA Athletics
Data Analyst

October 2021 – December 2022

- Built and managed an Azure database for athlete performance data, structuring longitudinal testing records for reliable ingestion, querying, and downstream analysis.
- Automated athlete profile refreshes and data monitoring workflows; developed interactive Power BI dashboards to visualize performance trends and review outliers.
- Applied time-series analysis to vertical jump measurements to identify performance degradation and injury-risk signals; translated analytical findings into recommendations that contributed to a significant reduction in injury rate.

ACADEMIC SERVICES

- Reviewer, IEEE Internet of Things (IoT) Journal, 2026
- Reviewer, Hawaii International Conference on System Sciences (HICSS), 2026
- Reviewer, Power Systems Computation Conference (PSCC), 2025
- Reviewer, Electric Power Systems Research (EPSR), 2025

AWARDS & HONORS

- NeurIPS Travel Award, 2025
- Seth Bonder Fellowship, 2025, 2026

TECHNICAL STRENGTHS

Programming Languages	Python, Julia, MATLAB, SQL, C++, R
Machine Learning	PyTorch, TensorFlow, scikit-learn, NumPy, pandas
Optimization & Solvers	JuMP, Gurobi, Ipopt, CPLEX, SCIP
Data & Visualization	Spark, Hadoop, PostgreSQL, Azure, Power BI, Matplotlib
Cloud/DevOps	Google Cloud, Kubernetes, Docker, Git, Bash
Software Engineering	MongoDB, Express, React, Node.js
Collaboration	Jira, Confluence